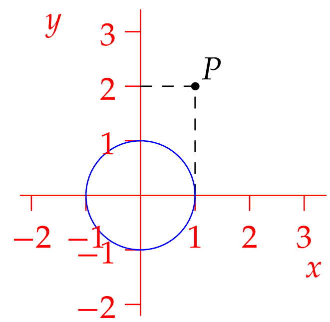
3 Analytic Geometry

Geometry in the style of Euclid and Hilbert is termed *synthetic*: axiomatic, without co-ordinates or explicit numerical measures of angle, length, area or volume. By contrast, the modern practice of geometry is typically *analytic*: reliant on algebra and co-ordinates (including vectors).

Analytic geometry arose in the early 1600s thanks to the work of René Descartes and Pierre de Fermat. The critical development was the *axis*[19] as a fixed reference ruler against which objects can be measured using *co-ordinates*. In honor of Descartes' work, the standard system within analytic geometry is called the Cartesian (literally *of Descartes*) co-ordinate system.

3.1 Cartesian Co-ordinates

Since Cartesian geometry should be very familiar, we merely sketch the core ideas.



- Perpendicular *axes* meet at the *origin* O. The axes are labelled using the real numbers.
- The *co-ordinates* of a point are measured by projecting orthogonally onto the axes. Since co-ordinates are pairs of real numbers, we denote the set of these

  $$\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$$

  In the picture, P has co-ordinates $(1, 2)$; usually written $P = (1, 2)$.
- Points may be manipulated algebraically via *addition* and *scalar multiplication*: if $P = (p_1, p_2)$ and $Q = (q_1, q_2)$, then

  $$P + Q := (p_1, p_2) + (q_1, q_2) = (p_1 + q_1, p_2 + q_2) \qquad \lambda P := (\lambda p_1, \lambda p_2)$$

- The *length* of a segment is computed using Pythagoras' Theorem

  $$d(P, Q) = |PQ| = \sqrt{(q_1 - p_1)^2 + (q_2 - p_2)^2}$$

  In the picture, $|OP| = \sqrt{1^2 + 2^2} = \sqrt{5}$. As in Section 2.5, segments are congruent if and only if they have the same length.
- *Curves* are defined using *equations*. For instance, $x^2 + y^2 = 1$ describes the pictured circle; by the distance formula, this is the locus of all points a distance 1 from the origin.

Analytic geometry was originally conceived as a computational toolkit built on top of Euclid's geometry. At first, mathematicians felt the need to justify analytic arguments synthetically lest no-one believe their work.[20] Synthetic geometry is not without its benefits, but its study has become a fringe activity in modern times; co-ordinates are just too useful to ignore!

[19] Originally there was only one axis, though a second was implied. Very quickly it became standard to have a second axis at right-angles to the first.

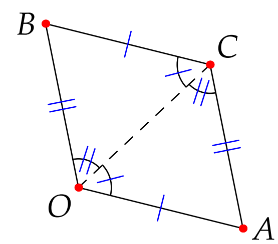
[20] This attitude persisted for some time. For example, the presentation in Issac Newton's groundbreaking *Principia* (1687) was largely synthetic, even though his private derivations made extensive use of co-ordinates and algebra.

It is important to get the history in the right order: in analytic arguments, we may assume anything from Euclid and mix strategies as appropriate. To see this at work, consider a simple result.

Lemma 3.1. *Suppose non-collinear points $O = (0,0)$, $A = (x,y)$ and $B = (v,w)$ are given and we define $C := (x+v, y+w)$. Then $OACB$ is a parallelogram.*

Proof. The distance formula shows that opposite sides have the same length

$$|BC| = \sqrt{x^2+y^2} = |OA|, \qquad |AC| = \sqrt{v^2+w^2} = |OB|$$



and therefore congruent.

SSS shows that $\triangle OAC \cong \triangle CBO$.

Euclid's discussion of alternate angles (pages 13–15) forces the opposite sides to be parallel. ■

Lemma 3.1 is essentially vector addition: feel free to use such notation/language if you prefer.

The next result is very useful: it gives an explicit parametrization for all points on a line through two given points.

Lemma 3.2. *The points X_t on the line $\overleftrightarrow{PQ}$ are in 1–1 correspondence with the real numbers via*

$$X_t = P + t(Q-P) = (1-t)P + tQ$$

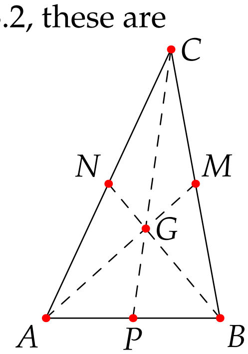
Moreover, $d(P, X_t) = |t|\,|PQ|$ so that t measures the (signed) distance along the line.

The proof is an exercise. As an example of how easy it can be to work in analytic geometry, we apply the Lemma to re-establish a famous result (compare Exercise 2.5.10c where we used Ceva's Theorem).

Theorem 3.3. *The medians of a triangle meet at a point $2/3$ of the way along each median.*

Proof. Given $\triangle ABC$, label the midpoints of each side as shown. By Lemma 3.2, these are

$$M = \frac{1}{2}(B+C), \quad N = \frac{1}{2}(A+B), \quad P = \frac{1}{2}(A+C)$$



The point $\frac{2}{3}$ of the way along median $\overline{AM}$ is then

$$G := A + \frac{2}{3}(M-A) = A + \frac{2}{3}(B+C-2A) = \frac{1}{3}(A+B+C)$$

By symmetry (check directly if you like), G is also $\frac{2}{3}$ of the way along the other two medians. ■

The analytic proof relied on adding points as abstract objects, though we could instead have expressed $A, B, C, \ldots$ in co-ordinates. Exercise 2 illustrates exactly this and, in an extension, of the biggest advantages of analytic geometry: the freedom to choose axes and co-ordinates so as to make calculations simple. This trick is essentially Euclid's superposition principle or Hilbert's congruence in disguise: we'll make this correspondence of concepts rigorous in shortly when we discuss *isometries*.

Exercises 3.1. *Key concepts: Analytic geometry is Euclidean geometry plus co-ordinates/algebra*

1. By completing the square, identify the curve described by the equation

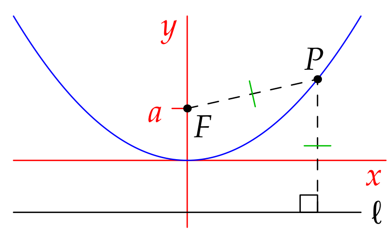
$$x^2 + y^2 - 4x + 2y = 10$$

2. (a) Perform a pure co-ordinate proof of Theorem 3.3. For simplicity, arrange the triangle so that $A = (0, 0)$ is the origin and B points along the positive x-axis.

   (b) Descartes and Fermat did not have a fixed perpendicular second axis. Their approach was equivalent to choosing a second axis oriented to make the problem as simple as possible.
   Given $\triangle ABC$, choose axes pointing along $\overline{AB}$ and $\overline{AC}$. Describe the co-ordinates of B and C with respect to such axes. Now give an even simpler proof of the centroid theorem (3.3).

3. Prove Lemma 3.2.

4. A *parabola* is a curve whose points are equidistant from a fixed point F (the *focus*) and a fixed line ℓ (the *directrix*).



   (a) Choose axes as shown in the picture so that $F = (0, a)$ and ℓ has equation $y = -a$. Find the equation of the parabola.

   (b) Now let e be a positive constant ($\neq 0, 1$). What is the equation of the parabola if ℓ has equation $y = -\frac{a}{e}$ and $\frac{|PF|}{|F\ell|} = e$? What happens in the limit $e \to 0^+$?